

Shor's Algorithm

Order finding, phase estimation, and the circuits that realise them

$$a \longmapsto U_a|_{\mathcal{H}_a} \cong S_r \longmapsto e^{2\pi i s/r} \xrightarrow{\text{QPE}} \frac{\ell}{Q} \approx \frac{s}{r} \xrightarrow{\text{CF}} r \longmapsto \gcd(a^{r/2} \pm 1, N)$$

The algorithm computes the multiplicative order of a random residue. The reduction of factoring to order finding, and the recovery of the order from the measurement, are classical. Running example throughout: $N = 21, a = 2, r = 6$.

Specification: order finding as stated in the literature

Inputs. (1) A black box $U_{x,N}$ performing $|j\rangle|k\rangle \mapsto |j\rangle|x^jk \bmod N\rangle$, for x coprime to the L -bit number N ; (2) $t = 2L + 1 + \lceil \log(2 + \frac{1}{2\epsilon}) \rceil$ qubits initialised to $|0\rangle$; (3) L qubits initialised to $|1\rangle$.

Outputs. The least integer $r > 0$ such that $x^r = 1 \pmod{N}$.

Runtime. $O(L^3)$ operations. Succeeds with probability $O(1)$.

Nielsen & Chuang, p. 232.

Three points to correct before using this as a contract.

1. $U_{x,N}$ is not free. Implementing controlled $U_x^{2^j}$ is the dominant cost of the algorithm.
2. One execution of the circuit does not return r . It returns a divisor of r ; see slide 5.
3. “Probability $O(1)$ ” is an upper bound and carries no information. The intended statement is $\Pr[\text{success}] = \Omega(1)$, and for the phase-estimation step the explicit constant is $4/\pi^2$.

The same algorithm, as a state evolution

Procedure (Nielsen & Chuang, p. 232; register order: counting $\otimes$ work).

- | | | |
|----|---|---|
| 1. | $ 0\rangle u\rangle$ | initial state |
| 2. | $\rightarrow \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} j\rangle 1\rangle$ | create superposition |
| 3. | $\rightarrow \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} j\rangle x^j \bmod N\rangle$ | apply $U_{x,N}$ |
| | $\approx \frac{1}{\sqrt{r}2^t} \sum_{s=0}^{r-1} \sum_{j=0}^{2^t-1} e^{2\pi i s j / r} j\rangle u_s\rangle$ | |
| 4. | $\rightarrow \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \widetilde{s/r}\rangle u_s\rangle$ | inverse Fourier transform on register 1 |
| 5. | $\rightarrow \widetilde{s/r}\rangle$ | measure register 1 |
| 6. | $\rightarrow r$ | continued-fraction algorithm |
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Two places where the $\approx$ and the arrow at step 6 hide something. Step 3: the identity $|x^j\rangle = r^{-1/2} \sum_s e^{2\pi i s j / r} |u_s\rangle$ is exact; the approximation sign belongs at step 4, where $\widetilde{s/r}$ denotes an estimate, exact only when $r \mid 2^t$. Step 6: continued fractions return $r / \gcd(s, r)$, so the arrow stands for the recovery loop of the previous slide.

Specification: SHORFACTOR

Precondition. $N \geq 3$ composite, odd, not a perfect power; $n = \lceil \log_2 N \rceil$.

Postcondition. Returns d with $1 < d < N$ and $d \mid N$.

Procedure (one attempt).

1. $a \leftarrow_{\$} \{2, \dots, N-1\}$; if $d = \gcd(a, N) > 1$ return d .
2. $r \leftarrow \text{ORDERFIND}(N, a)$.
3. If r even and $a^{r/2} \not\equiv -1$: return $\gcd(a^{r/2} - 1, N)$.
4. Else draw a new a .

(quantum)

Correctness. Step 3's hypothesis holds with probability $\geq \frac{1}{2}$ over a . Every candidate is checked by division: the algorithm is Las Vegas. Failure costs one repetition.

Complexity. $\tilde{O}(n^3)$ quantum operations, $\Theta(n)$ logical qubits, expected $O(1)$ attempts; classical post-processing $O(n^3)$ bit operations.

Specification: ORDERFIND, one circuit execution

Precondition. $\gcd(a, N) = 1$; $r = \text{ord}_N(a) < N$; $t = \lceil 2 \log_2 N \rceil$, $Q = 2^t \geq N^2$; counting register $|0\rangle^{\otimes t}$, work register $|1\rangle_n$, all ancillas $|0\rangle$.

Available operation. A reversible circuit for controlled multiplication by the classically precomputed constant $a^{2^j} \bmod N$. Not an oracle.

Postcondition. The measured ℓ is distributed as though s were drawn uniformly from $\{0, \dots, r-1\}$ and phase estimation run on s/r . Conditioned on s ,

$$\Pr \left[\left| \frac{\ell}{Q} - \frac{s}{r} \right| \leq \frac{1}{2Q} < \frac{1}{2N^2} \right] \geq \frac{4}{\pi^2} = 0.405 \dots \quad (\text{distance mod } 1)$$

Continued fractions then return $q = r / \gcd(s, r)$, a *divisor* of r ; $q = r$ iff $\gcd(s, r) = 1$. All ancillas return to $|0\rangle$.

Recovery. $R \leftarrow \text{lcm}(q_1, q_2, \dots)$ over independent executions until $a^R \equiv 1$; then $R = r$. Two executions suffice with probability > 0.54 .

Complexity. $\Theta(n^3)$ Toffoli gates, $\Theta(n)$ logical qubits.

Plan

1. Factoring reduces to order finding. Classical.
2. The postulates used, and where each is invoked.
3. U_a on the orbit subspace is a cyclic shift; its eigenvalues are $e^{2\pi is/r}$.
4. Phase kickback and phase estimation; the inverse Fourier transform.
5. Why the input $|1\rangle$ suffices; the order-finding circuit.
6. Continued fractions; why $t \simeq 2n$.
7. One complete run at $N = 21$.
8. Where the cost lies.
9. The same machine on the elliptic-curve discrete logarithm.
10. Comparison with the classical algorithms.

Each section opens with the obstacle it removes, set off by a rule in the margin. Postulates are labelled **P1**–**P5** on slide 9 and the badge is repeated at each point of use.

1. Factoring reduces to order finding

| *N resists direct attack. But a non-trivial square root of 1 splits it, and orders manufacture such roots.*

Lemma. If $x^2 \equiv 1 \pmod{N}$ and $x \not\equiv \pm 1$, then $\gcd(x-1, N)$ is a non-trivial factor.

Proof. $N \mid (x-1)(x+1)$. If $\gcd(x-1, N) = N$ then $x \equiv 1$; if $\gcd(x-1, N) = 1$ then $N \mid x+1$, so $x \equiv -1$. Both excluded, hence $1 < \gcd(x-1, N) < N$. □

Construction of such an x . Let $r = \text{ord}_N(a)$. If r is even put $x = a^{r/2}$. Then $x^2 = a^r \equiv 1$, and $x \not\equiv 1$ since $r/2 < r$ would contradict minimality of r . The remaining condition is $a^{r/2} \not\equiv -1$.

Theorem. For odd composite N with $m \geq 2$ distinct prime factors and a uniform in $(\mathbb{Z}/N\mathbb{Z})^\times$,

$$\Pr [r \text{ even and } a^{r/2} \not\equiv -1] \geq 1 - 2^{-(m-1)} \geq \frac{1}{2}.$$

Proof by CRT on the 2-adic valuations $v_2(\text{ord}_{p_i^{a_i}}(a_i))$: the bad event forces them all equal, and each takes any fixed value with probability $\leq \frac{1}{2}$. The hypotheses “ N odd” and “ $m \geq 2$ ” are both used, which is why even N and perfect powers are removed first.

Running example: $N = 21, a = 2$

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 11 \rightarrow 1 \pmod{21}, \quad r = \text{ord}_{21}(2) = 6.$$

$r = 6$ is even and $2^3 = 8 \not\equiv -1 \equiv 20$, so

$$\gcd(2^3 - 1, 21) = \gcd(7, 21) = 7, \quad \gcd(2^3 + 1, 21) = \gcd(9, 21) = 3.$$

No quantum computation has occurred. The only quantity supplied by the quantum circuit is $r = 6$.

$N = 21$ rather than $N = 15$: every order modulo 15 divides 4, so $Q \mid$ the period, phase estimation is exact, and the continued-fraction step is vacuous. With $r = 6 \nmid Q = 2^t$ both steps are exercised.

2. The postulates used

Everything that follows is assembled from four rules and one corollary. Nothing else is assumed.

- (P1) State** A state is a unit vector $|\psi\rangle$ in $\mathbb{C}^{2^k}$, determined up to a global phase. A register stores an integer: $|x\rangle = |x_1\rangle \otimes \cdots \otimes |x_k\rangle$, $x = \sum_j x_j 2^{k-j}$.
- (P2) Evolution** Gates are unitary: $U^\dagger U = U U^\dagger = \mathbb{1}$. Hence invertible, hence every subroutine must be written reversibly.
- (P3) Measurement** Measuring in the computational basis returns x with probability $|\langle x|\psi\rangle|^2$. This is the only channel from amplitudes to classical data.
- (P4) Composition** Registers combine by tensor product; $t + n$ qubits carry 2^{t+n} amplitudes.
- (P5) Interference** Amplitudes add before they are squared: if x is reachable along several computational paths, $\Pr[x] = |\sum_j A_j|^2$, whereas the classical value is $\sum_j |A_j|^2$. The cross terms permit cancellation.

Consequence of **(P5)**: if any register retains a record $|g_j\rangle$ of which path was taken, the cross term for paths $j \neq j'$ acquires the factor $\langle g_{j'}|g_j\rangle$ and vanishes for orthogonal records. Branch-dependent scratch data must therefore be uncomputed before the branches are recombined.

Two consequences of **P2** used repeatedly

- (a) A unitary preserves norms and inner products, and is exactly a map carrying an orthonormal basis to an orthonormal basis. A permutation matrix is therefore unitary.
- (b) A unitary U has an orthonormal eigenbasis, and every eigenvalue satisfies $|\lambda| = 1$:

$$\| |u\rangle \| = \| U|u\rangle \| = |\lambda| \| |u\rangle \|.$$

So each eigenvalue is $e^{2\pi i\theta}$ for a unique $\theta \in [0, 1)$, and $U^m |u\rangle = e^{2\pi i m\theta} |u\rangle$.

An eigenvalue of a unitary is a phase. Section 3 puts the order r into such a phase; section 4 measures it.

3. Modular multiplication as a matrix

The orbit of a has length up to N , so it cannot be walked. Look at the spectrum of an operator instead — and the operator has to be unitary.

Let $n = \lceil \log_2 N \rceil$. On the n -qubit work register define

$$U_a|y\rangle := \begin{cases} |ay \bmod N\rangle, & 0 \leq y < N, \\ |y\rangle, & N \leq y < 2^n. \end{cases}$$

Unitarity **(P2)**. On $\{0, \dots, N-1\}$ the map $y \mapsto ay$ is injective, since a is invertible mod N , hence bijective on a finite set; on $\{N, \dots, 2^n-1\}$ it is the identity. So U_a permutes the computational basis.

The second branch is required: the register holds 2^n states while multiplication acts on N of them. Without the padding the operator is not a permutation of the basis and not unitary.

Powers. $U_a U_b = U_{ab \bmod N}$, hence

$$U_a^{2^j} = U_{a^{2^j} \bmod N}$$

with the constants $a^{2^j} \bmod N$ obtained by j classical squarings. Each controlled power therefore costs one modular multiplication rather than 2^j .

The orbit subspace

$$\mathcal{O}_a := \{a^k \bmod N : 0 \leq k < r\}, \quad \mathcal{H}_a := \text{span}\{|z\rangle : z \in \mathcal{O}_a\} \subseteq \mathbb{C}^{2^n}.$$

Proposition. $|\mathcal{O}_a| = r$, $U_a(\mathcal{H}_a) = \mathcal{H}_a$, and under $V|a^k \bmod N\rangle := |k\rangle$ ($k \in \mathbb{Z}/r\mathbb{Z}$),

$$V(U_a|_{\mathcal{H}_a})V^\dagger = S_r, \quad S_r|k\rangle = |k+1 \bmod r\rangle.$$

Proof. If $a^k \equiv a^{k'}$ with $0 \leq k < k' < r$ then $a^{k'-k} \equiv 1$ with $0 < k' - k < r$, contradicting minimality of r ; so the r elements are distinct and V is unitary. Since $a^r \equiv 1$ the index is well defined mod r , and $VU_aV^\dagger|k\rangle = VU_a|a^k\rangle = V|a^{k+1}\rangle = |k+1 \bmod r\rangle$. $\square$

Modular multiplication, restricted to the orbit of 1, is the cyclic shift on $\mathbb{Z}/r\mathbb{Z}$.

The Fourier matrix diagonalises the shift

Let $\omega_r = e^{2\pi i/r}$ and $(F_r)_{s,k} = \omega_r^{sk} / \sqrt{r}$. Then

$$F_r S_r F_r^\dagger = \text{diag}(1, \omega_r, \omega_r^2, \dots, \omega_r^{r-1}).$$

Proof. $(F_r S_r)_{s,k} = (F_r)_{s,k+1} = \omega_r^{s(k+1)} / \sqrt{r} = \omega_r^s (F_r)_{s,k} = (D F_r)_{s,k}$ with $D = \text{diag}(\omega_r^s)$.
Hence $F_r S_r = D F_r$, and F_r is unitary by $\sum_k \omega_r^{sk} \overline{\omega_r^{s'k}} = r \delta_{ss'}$. □

Why roots of unity. Solving $S_r v = \lambda v$ directly: $v_k = \lambda^{-k} v_0$, and closing the cycle, $v_0 = v_r = \lambda^{-r} v_0$, forces

$$\lambda^r = 1.$$

The eigenvalues are the r -th roots of unity because the index set is $\mathbb{Z}/r\mathbb{Z}$.

The same change of basis diagonalises every matrix commuting with S_r , i.e. every circulant. It is the transform behind the NTT, where it converts convolution into coordinatewise multiplication.

Eigenvalues of U_a on the orbit subspace

Theorem. For $s = 0, \dots, r-1$ set

$$|u_s\rangle := \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |a^k \bmod N\rangle \in \mathcal{H}_a.$$

Then $U_a |u_s\rangle = e^{2\pi i s / r} |u_s\rangle$. The r eigenvalues are distinct, so $|u_0\rangle, \dots, |u_{r-1}\rangle$ is an orthonormal eigenbasis of $\mathcal{H}_a$.

Proof. Shift the summation index, using $a^r \equiv 1$:

$$U_a |u_s\rangle = \frac{1}{\sqrt{r}} \sum_k e^{-2\pi i s k / r} |a^{k+1}\rangle = \frac{1}{\sqrt{r}} \sum_{k'} e^{-2\pi i s (k'-1) / r} |a^{k'}\rangle = e^{2\pi i s / r} |u_s\rangle. \quad \square$$

The period r is now the denominator of an eigenvalue phase $\theta_s = s/r$. Order finding has become phase estimation.

$N = 21, a = 2$: $\mathcal{O}_2 = \{1, 2, 4, 8, 16, 11\}$, and the eigenvalues of U_2 on $\mathcal{H}_2$ are the sixth roots of unity, phases $\{0, \frac{1}{6}, \frac{2}{6}, \frac{3}{6}, \frac{4}{6}, \frac{5}{6}\}$.

4. Phase kickback

r now sits in an eigenvalue, but an eigenvalue on its eigenvector is a global phase, and by (P1) those are unobservable.

A controlled gate makes it relative.

Lemma. If $U|u\rangle = e^{2\pi i\theta}|u\rangle$ then, for all α, β ,

$$c\text{-}U[(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle] = (\alpha|0\rangle + \beta e^{2\pi i\theta}|1\rangle) \otimes |u\rangle.$$

Proof. $c\text{-}U|c\rangle|\psi\rangle = |c\rangle U^c|\psi\rangle$; by linearity the left side is $\alpha|0\rangle U^0|u\rangle + \beta|1\rangle U^1|u\rangle$. The scalar $e^{2\pi i\theta}$ commutes with the tensor factors (P4), so it may be attached to the control branch and $|u\rangle$ factored out. $\square$

The eigenvalue was already a property of the pair $(U, |u\rangle)$. What changes is its position: a relative phase between the two control branches, which (P5) makes observable. The target is returned unchanged and can drive further gates.

The controlled-power ladder

Prepare t counting qubits in $|0\rangle^{\otimes t}$, apply $H^{\otimes t}$ **(P4)**, then $c\text{-}U^{2^{t-j}}$ from qubit j , $j = 1, \dots, t$.
Writing $Q = 2^t$,

$$\bigotimes_{j=1}^t \frac{|0\rangle + e^{2\pi i 2^{t-j}\theta} |1\rangle}{\sqrt{2}} \otimes |u\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} e^{2\pi i x \theta} |x\rangle \otimes |u\rangle.$$

Proof. $U^m|u\rangle = e^{2\pi i m\theta}|u\rangle$, so the lemma places $e^{2\pi i 2^{t-j}\theta}$ on the $|1\rangle$ branch of qubit j .
Expanding the product over $x = \sum_j x_j 2^{t-j}$, $\prod_j e^{2\pi i x_j 2^{t-j}\theta} = e^{2\pi i x \theta}$. □

Two remarks.

- By **(P3)**, measuring here returns nothing: every amplitude has modulus $Q^{-1/2}$, so all Q outcomes are equiprobable. The information is in relative phases.
- $U_a^{2^{t-j}} = U_{a^{2^{t-j} \bmod N}}$, so each rung is one modular multiplication. Without this the ladder would contain 2^t operations. This is repeated squaring, used by classical modular exponentiation for the same reason.

The inverse Fourier transform converts phase into position

With $\omega_Q = e^{2\pi i/Q}$ and $(F_Q)_{y,x} = \omega_Q^{yx} / \sqrt{Q}$, the circuit applies $F_Q^\dagger$. Since F_Q diagonalises the shift on $\mathbb{Z}/Q\mathbb{Z}$, $F_Q^\dagger$ maps eigenvalue phases back to positions. Explicitly it computes, for each label ℓ ,

$$\alpha_\ell = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x \theta} e^{-2\pi i x \ell / Q} = \frac{1}{Q} \sum_{x=0}^{Q-1} e^{2\pi i x \delta_\ell}, \quad \delta_\ell := \theta - \frac{\ell}{Q}.$$

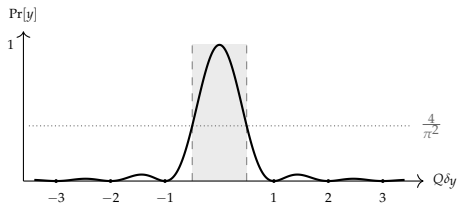
A sum of Q unit vectors whose heading advances by $2\pi\delta_\ell$ per step **P5**:

- $\delta_\ell = 0$: all terms equal 1, $\alpha_\ell = 1$.
- $\delta_\ell = k/Q, k \neq 0$: the terms are the Q -th roots of unity taken k at a time, a closed polygon, $\alpha_\ell = 0$ exactly.
- $|\delta_\ell|$ small: the terms form a short arc and the sum remains large.

Summing the geometric series,

$$\alpha_\ell = e^{\pi i(Q-1)\delta_\ell} \frac{\sin(\pi Q \delta_\ell)}{Q \sin(\pi \delta_\ell)}, \quad \Pr[\ell] = \frac{\sin^2(\pi Q \delta_\ell)}{Q^2 \sin^2(\pi \delta_\ell)}.$$

The output distribution



Theorem. Let b be an integer nearest $Q\theta$ mod Q , so $|\delta_b| \leq \frac{1}{2Q}$. Then

$$\Pr[b] \geq \frac{4}{\pi^2} = 0.4052 \dots$$

Proof. With $\vartheta = \pi Q|\delta_b| \in (0, \pi/2]$,
 $\sqrt{\Pr[b]} = \frac{1}{Q} \frac{\sin \vartheta}{|\sin(\pi\delta_b)|}$. Bound
 $|\sin(\pi\delta_b)| \leq \pi|\delta_b|$ and, by Jordan's inequality,
 $\sin \vartheta \geq \frac{2}{\pi}\vartheta = 2Q|\delta_b|$. Hence $\sqrt{\Pr[b]} \geq 2/\pi$.
 $\square$

The two grid points straddling $Q\theta$ carry
 $\geq 8/\pi^2 = 0.81$. Both bounds are tight.

Since $4/\pi^2 < \frac{1}{2}$, majority amplification of the nearest-point event does not apply. Accuracy is obtained from additional counting qubits, at cost $\lceil \log_2(2 + \frac{1}{2\epsilon}) \rceil$ for failure probability ϵ . Repetition is used differently, on slide 21.

5. The input $|1\rangle$ suffices

Phase estimation wants an eigenvector as input, and writing down $|u_s\rangle$ requires r — the quantity being computed.

Theorem. $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$.

Proof. Substituting and exchanging the order of summation,

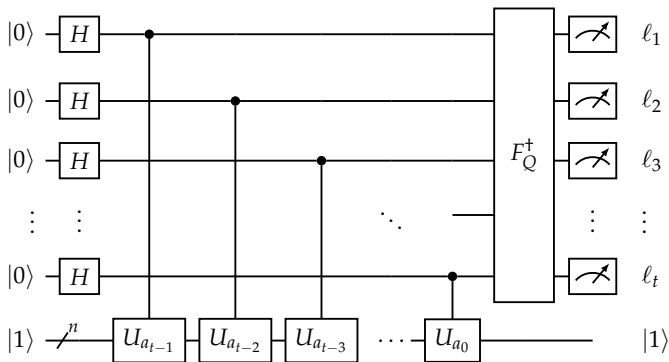
$$\frac{1}{\sqrt{r}} \sum_s |u_s\rangle = \frac{1}{r} \sum_k \left(\sum_s e^{-2\pi i s k / r} \right) |a^k\rangle = \frac{1}{r} \cdot r \delta_{k,0} |a^0\rangle = |1\rangle,$$

by $\sum_{s=0}^{r-1} \omega_r^{-sk} = r [k \equiv 0]$. □

The circuit is linear (P2), and a measurement of the work register in $\{|u_s\rangle\}$ commutes with every gate acting on it. Hence the counting-register statistics are those obtained by drawing s uniformly from $\{0, \dots, r-1\}$ and running phase estimation on $\theta_s = s/r$.

$|1\rangle = |0 \cdots 01\rangle$ costs one X gate. The uniform numerator s is obtained without knowing r .

The order-finding circuit



$a_j := a^{2^j} \bmod N$, precomputed by j classical squarings. The state before $F_Q^\dagger$ has two equal descriptions,

$$\underbrace{\frac{1}{\sqrt{Q}} \sum_x |x\rangle |a^x \bmod N\rangle}_{\text{periodic function}} = \underbrace{\frac{1}{\sqrt{r}} \sum_s \left(\frac{1}{\sqrt{Q}} \sum_x e^{2\pi i x s / r} |x\rangle \right) |u_s\rangle}_{\text{eigenphase branches}},$$

the first Shor's 1994 formulation, the second phase estimation. The work register is not measured; measuring it and discarding the outcome leaves the counting-register marginal unchanged **P3**.

6. From ℓ to r

| The measurement gives ℓ with ℓ/Q only near s/r , and infinitely many rationals lie near any real number.

Bounding the denominator makes the rational unique.

Theorem (Legendre). If p/q is in lowest terms, $q \geq 1$, and $|x - \frac{p}{q}| < \frac{1}{2q^2}$, then p/q is a convergent of the continued-fraction expansion of x .

Corollary. The denominator sought is at most $r < N$, and Legendre's threshold scales as q^{-2} . So it suffices that

$$\left| \frac{\ell}{Q} - \frac{s}{r} \right| < \frac{1}{2N^2},$$

and phase estimation delivers $|\ell/Q - s/r| \leq \frac{1}{2Q}$. Hence

$$Q \geq N^2, \quad \text{that is,} \quad t \geq 2n.$$

This inequality determines the size of the counting register, and therefore the $2n$ modular multiplications that dominate the circuit. The constant $\frac{1}{2}$ cannot be improved to 1: for $x = 0.4$ and $p/q = 1/3$ the weaker bound holds, yet $1/3$ is not a convergent of $2/5$.

What one execution returns

Continued fractions return s/r in lowest terms, so the denominator obtained is

$$q = \frac{r}{\gcd(s, r)} \mid r,$$

equal to r exactly when $\gcd(s, r) = 1$, which occurs with probability $\varphi(r)/r$.

Recovery. Over independent executions accumulate

$$R_m := \text{lcm}(q_1, \dots, q_m)$$

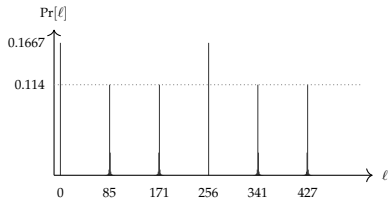
and stop when $a^{R_m} \equiv 1 \pmod{N}$; then $R_m = r$. Two executions suffice whenever $\gcd(s_1, s_2) = 1$, which holds with probability $\geq 1 - \sum_p p^{-2} = 1 - 0.4522\dots > 0.54$.

The test $a^R \equiv 1$ is a classical verification, and the extracted factor is checked by division. The procedure is Las Vegas: an unlucky sample, or a noisy execution, costs one repetition and cannot produce an incorrect factorisation.

Testing small multiples $2q, 3q, \dots$ of a single q is a practical shortcut but not a worst-case repair, since $\lfloor N/q \rfloor$ may be exponential in n .

7. One complete run: $N = 21, a = 2$

Every piece above, assembled on one instance, with exact numbers.



Exact distribution, $t = 9, Q = 512$.

$n = 5$, and $Q \geq N^2 = 441$ forces $Q = 512, t = 9$.

Six peaks at $\ell \approx 512s/6$, i.e.

$\{0, 85, 171, 256, 341, 427\}$, carrying 78.9% of the mass. Branches $s = 0, 3$ lie on exact grid points; the remaining four place 0.114 on the nearest point, i.e. 0.684 within their branch.

Peak heights are 0.16667 rather than exactly $1/6$: $6 \nmid 512$, since $512 = 6 \cdot 85 + 2$, so two residue classes of x contain 86 values and four contain 85.

$$\frac{85}{512} = [0; 6, 42, 2] \Rightarrow \text{convergents } 0, \frac{1}{6}, \dots \Rightarrow q = 6, \quad 2^6 = 64 \equiv 1 \checkmark$$

$$\frac{171}{512} \Rightarrow \frac{1}{3} \Rightarrow q = 3; \quad \frac{256}{512} = \frac{1}{2} \Rightarrow q = 2; \quad \text{lcm}(3, 2) = 6, \quad 2^6 \equiv 1 \checkmark$$

Then $\gcd(2^3 - 1, 21) = 7$ and $\gcd(2^3 + 1, 21) = 3$. One execution returns $r = 6$ directly with probability 0.3208; $\ell = 0$ carries no information.

8. Where the cost lies

After all that Fourier analysis, the natural guess is that the transform dominates. It is not close.

Stage	Cost	Note
$H^{\otimes t}$	$t = 2n + O(1)$ Clifford gates	negligible
modular exponentiation	$\Theta(n^3)$ Toffoli; $\approx 0.3n^3$ windowed	$2n$ mult. $\times n$ add. $\times \Theta(n)$
$F_Q^\dagger$	$\Theta(n^2)$ exact, $O(n \log n)$ approximate	Coppersmith
continued fractions	$O(n^3)$ bit operations	classical
total	$\Theta(n)$ logical qubits, $\Theta(n^3)$ Toffoli	$T \approx 4 \times$ Toffoli

At $n = 2048$: modular exponentiation $\approx 2.6 \times 10^9$ Toffoli against $\approx 9 \times 10^4$ gates for the approximate transform, a ratio $\approx 3 \times 10^4$. Asymptotically $\Theta(n^3)$ against $O(n \log n)$. The dominant cost is reversible modular arithmetic.

	Beauregard 2003	Gidney–Ekerå 2019/21	Gidney 2025
logical qubits	$2n + 3$	$\approx 3n$ (≈ 6200)	≈ 1400
gate count	$O(n^3 \log n)$ elem.	$\approx 3 \times 10^9$ Toffoli	6.5×10^9 Tof/run, ≈ 9 runs
physical qubits	—	20 million	$\approx 9 \times 10^5$
runtime	—	8 hours	≈ 5 days

Beauregard’s figure counts elementary gates, including rotations, and is not comparable Toffoli-for-Toffoli. Physical estimates assume a 10^{-3} gate error rate, $1 \mu\text{s}$ cycle time, $10 \mu\text{s}$ reaction time. Status of these figures: August 2026.

9. The elliptic-curve discrete logarithm

The same machine on a different group — and one structural change that replaces continued fractions by a single inversion.

Instance. $E(\mathbb{F}_p)$ with $p > 3$ prime, a point $P \in E(\mathbb{F}_p)$ of prime order r , and $Q = [m]P$; recover m . Put $n = \lceil \log_2 p \rceil$. Hasse's bound $|\#E(\mathbb{F}_p) - (p + 1)| \leq 2\sqrt{p}$ gives $r \leq \#E(\mathbb{F}_p) < 2^{n+1}$, so $n + 1$ qubits hold an index.

The group and the oracle. Take $A = \mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$ and

$$f(k, \ell) := [k]P + [\ell]Q = [k + m\ell]P.$$

Then $f(k, \ell) = f(k', \ell')$ exactly when $k + m\ell \equiv k' + m\ell' \pmod{r}$, so f is constant on the cosets of

$$L := \{(-mt, t) : t \in \mathbb{Z}/r\mathbb{Z}\} = \langle (-m, 1) \rangle \leq A$$

and separates distinct cosets. Only the group law on $E(\mathbb{F}_p)$ was used, so the same statement holds for $\mathbb{F}_p^\times$ with $f(k, \ell) = g^k h^\ell \pmod{p}$.

L is a line through the origin in A , of known size r and unknown slope. Order finding is the rank-1 instance of the same problem: there a subgroup of $\mathbb{Z}/Q\mathbb{Z}$ is determined by its size alone, and the size is the unknown. The two unknowns dictate the two post-processing steps, continued fractions against one inversion.

The measured relation, derived

Two index registers of dimension r , a point register in $|\mathcal{O}\rangle$, and $\omega_r = e^{2\pi i/r}$. All sums run over $\mathbb{Z}/r\mathbb{Z}$.

(1) **Superpose and query** P1 P4 $\frac{1}{r} \sum_{k,\ell} |k\rangle |\ell\rangle |\mathcal{O}\rangle \xrightarrow{f} \frac{1}{r} \sum_{k,\ell} |k\rangle |\ell\rangle |[k + m\ell]P\rangle.$

(2) **Discard the point register** P3 It holds $[c]P$ for a uniform c , so the index registers collapse to the uniform superposition over the coset $\{(k, \ell) : k + m\ell \equiv c\}$, which the free variable ℓ parametrises: $\frac{1}{\sqrt{r}} \sum_{\ell} |c - m\ell\rangle |\ell\rangle.$

(3) **Apply $F_r^\dagger$ to both index registers** P5

$$\frac{1}{r^{3/2}} \sum_{y_1, y_2} \omega_r^{-cy_1} \underbrace{\sum_{\ell} \omega_r^{\ell(my_1 - y_2)}}_{= r[y_2 \equiv my_1]} |y_1\rangle |y_2\rangle = \frac{1}{\sqrt{r}} \sum_{y_1} \omega_r^{-cy_1} |y_1\rangle |my_1 \bmod r\rangle,$$

by the geometric-sum lemma of section 3, applied in the variable ℓ .

(4) **Measure** P3 The outcome is $(y_1, my_1 \bmod r)$ with y_1 uniform on $\mathbb{Z}/r\mathbb{Z}$.

$\omega_r^{-cy_1}$ varies with y_1 , so it is not a global phase — but it sits on the basis states of the register measured immediately after, and its modulus is 1 for every c . The unknown coset label does not enter the statistics, so the point register need never be measured.

Recovering m

Corollary. If $\gcd(y_1, r) = 1$ then $m \equiv y_2 y_1^{-1} \pmod{r}$, by one extended-Euclid inversion. For prime r — the NIST curves have cofactor 1 — every $y_1 \neq 0$ is invertible, so one execution succeeds with probability $(r - 1)/r$. The outcome $y_1 = 0$ names the trivial character, which annihilates every subgroup and therefore carries no information about m .

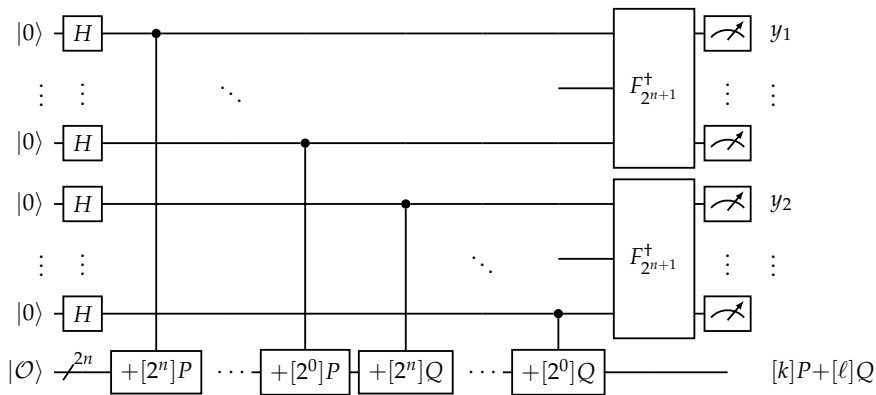
Finite registers. r is not a power of two, so each index register carries $n + 1$ qubits and one applies $F_{2^{n+1}}^\dagger$. The exact delta function broadens into the Fejér peak of section 4, and the measured pair satisfies

$$\frac{y_1}{2^{n+1}} \approx \frac{j_1}{r}, \quad \frac{y_2}{2^{n+1}} \approx \frac{j_2}{r}, \quad j_2 \equiv m j_1 \pmod{r}.$$

Rounding each ratio to the nearest rational with denominator r recovers (j_1, j_2) . The denominator is known here, so this is a rounding in two coordinates and no search over denominators occurs.

The relation obeyed by measured data is the post-transform $y_2 \equiv m y_1$. The pre-transform relation $k + m\ell \equiv c$ constrains registers that are never measured; applying it to the outcomes, as in $\hat{m} = (c - k)\ell^{-1}$, returns the $1/r$ random baseline even on an exact simulator. Verify $[\hat{m}]P = Q$ before accepting, which makes the ECDLP procedure Las Vegas as well.

The circuit



Top three wires: the k -register, $n + 1$ qubits. Middle three: the ℓ -register. Bottom: the point register, $2n$ qubits for the affine coordinates.

Double scalar multiplication. Since $[s]R = \sum_j s_j [2^j]R$, the ladder of controlled additions of the classical constants $[2^j]R$ realises $|s\rangle|O\rangle \mapsto |s\rangle|[s]R\rangle$; with $R = P$ on the k -register and $R = Q$ on the ℓ -register this produces $|k\rangle|\ell\rangle|[k]P + [\ell]Q\rangle$. The constants come from j classical doublings.

Order finding and the ECDLP side by side

	order finding	ECDLP
group A	$\mathbb{Z}/Q\mathbb{Z}$	$\mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$
hidden subgroup L	$r\mathbb{Z}/Q\mathbb{Z}$	$\langle(-m, 1)\rangle$
rank of A	1	2
transforms	one F_Q^+	two $F_{2^{n+1}}^+$
unknown	the size of L	the slope of L
one outcome	$\ell \approx Qs/r$	(y_1, y_2) with $y_2 \equiv my_1$
post-processing	continued fractions	one modular inversion
success per execution	$\varphi(r)/r$	$(r-1)/r$ for prime r
oracle	$x \mapsto a^x \bmod N$	$(k, \ell) \mapsto [k]P + [\ell]Q$

Only cyclicity of the group and an efficient reversible circuit for its operation entered the derivation, so replacing the oracle by $(k, \ell) \mapsto g^k h^\ell \bmod p$ gives the finite-field discrete logarithm with no other change. RSA, finite-field Diffie–Hellman, DSA, ECDH and ECDSA are therefore four deployed families broken by one mechanism, so the migration to post-quantum primitives is a single migration covering all four.

In the ECDLP column every box adds a point known at compile time, so a constant control deletes or fixes a Toffoli and no quantum–quantum point addition occurs anywhere; the additions commute, so their order is free.

Elliptic curves against RSA

The irreducible cost inside a point addition is the modular inversion in $\lambda = (y_2 - y_1)/(x_2 - x_1)$. One controlled addition contains 4 inversions, 2 squarings and 4 multiplications, giving the leading coefficient $4 \cdot 32 + 2 \cdot 16 + 4 \cdot 16 = 224$, and a run performs $2n$ of them sequentially:

$$9n + 2\lceil \log_2 n \rceil + 10 \text{ logical qubits,} \quad (448 \log_2 n + 4090) n^3 \text{ Toffoli.}$$

	ECDLP, P-256	RSA-3072	ratio
classical security	≈ 128 bit	≈ 128 bit	matched
operand size n	256	3 072	$12\times$
logical qubits	2 330	6 146	$0.38\times$
Toffoli gates	1.26×10^{11}	1.86×10^{13}	$0.007\times$

At matched classical security a 256-bit curve pairs with a 3072-bit modulus, and both quantum costs are cubic in the operand size. The $12\times$ smaller operand supplies a factor $12^3 \approx 1.7 \times 10^3$; the inversion inside each point addition returns about one order of magnitude of it, leaving the observed $\approx 1.5 \times 10^2$. Elliptic curves owe their classical economy to the small operand, and the same small operand leaves them the cheaper quantum target.

Simulated logical counts from minimal-width constructions on both sides, so the ratio is like-for-like; the windowed $\approx 0.3n^3$ construction of section 8 is a different design point. Ekerå–Håstad divides the RSA column by 4 and does not apply to a general ECDLP. Full tables and caveats in the backup.

10. Against the classical algorithms

Against what, classically?

Factoring and the finite-field DLP. The general number field sieve runs in heuristic time

$$L_N\left[\frac{1}{3}, c\right] = \exp\left((c + o(1))(\ln N)^{1/3}(\ln \ln N)^{2/3}\right), \quad c = (64/9)^{1/3} = 1.9229\dots,$$

which in the bit length $n = \log_2 N$ reads $\exp(O(n^{1/3} \log^{2/3} n))$: it exceeds every polynomial in n and is exceeded by every $2^{\epsilon n}$. Against Shor's $\Theta(n^3)$ the separation is super-polynomial.

ECDLP. On a curve of prime order r there is no index calculus. The best known attack is generic — Pollard's ρ , expected $\sqrt{\pi r/4} = \Theta(2^{n/2})$ — and Shoup's $\Omega(\sqrt{r})$ bound holds in the generic group model. Against $\Theta(n^3 \log n)$ the separation is exponential.

	best classical	quantum	at 128-bit security
factoring	$\exp(O(n^{1/3} \log^{2/3} n))$	$\Theta(n^3)$ Toffoli	2^{128} vs $\approx 2^{44}$
ECDLP	$\Theta(2^{n/2})$	$\Theta(n^3 \log n)$ Toffoli	2^{128} vs $\approx 2^{37}$

Consequence for key sizes. Inverting the sieve, an RSA modulus at σ bits of classical security needs $n = \Theta(\sigma^3 / \log^2 \sigma)$ and so a quantum cost $\Theta(\sigma^9 / \log^6 \sigma)$; a curve needs $n = 2\sigma$ and so a quantum cost $\Theta(\sigma^3)$. Raising the classical security level pushes RSA out of quantum reach much faster than it does ECC.

The two columns count different objects — classical operations against logical Toffoli gates — and the quantum column carries an error-correction multiplier. The asymptotic separation survives both; a crossover estimate at one key size is far more fragile.

Summary

1. Factoring reduces to order finding: $\gcd(a^{r/2} \mp 1, N)$ splits N when r is even and $a^{r/2} \not\equiv -1$, which holds with probability $\geq \frac{1}{2}$ over a .
2. U_a restricted to the orbit of 1 is the cyclic shift S_r ; the Fourier matrix diagonalises S_r ; hence the eigenvalues of U_a there are $e^{2\pi i s/r}$.
3. A controlled- U ladder writes θ as a phase ramp; $F_Q^\dagger$ converts the ramp into a position, obtained within $1/(2Q)$ of Qs/r with probability $\geq 4/\pi^2$.
4. $|1\rangle$ is the uniform superposition of all r eigenvectors, so the run samples s uniformly, and Legendre's theorem forces $Q \geq N^2$, hence $t \simeq 2n$ and $2n$ modular multiplications.
5. One execution returns a divisor of r ; lcm over executions plus the test $a^R \equiv 1$ returns r , so the procedure is Las Vegas. The dominant cost is reversible modular arithmetic, the Fourier transform a lower-order term.
6. The same circuit computes discrete logarithms: the ECDLP is the rank-2 instance on $\mathbb{Z}/r\mathbb{Z} \times \mathbb{Z}/r\mathbb{Z}$ with hidden subgroup $\langle(-m, 1)\rangle$ — two transforms, one modular inversion, success $(r-1)/r$, and $(448 \log_2 n + 4090)n^3$ Toffoli on $9n + 2\lceil \log_2 n \rceil + 10$ qubits.
7. Classically the sieve is sub-exponential, $\exp(O(n^{1/3} \log^{2/3} n))$, and the generic ECDLP attack is exponential, $\Theta(2^{n/2})$; both quantum costs are $\tilde{\Theta}(n^3)$.

Every step used that the group is abelian, hence that its irreducible representations have dimension one, hence that each eigenvalue is a single phase. Over a non-abelian group an eigenvalue becomes a matrix and the kickback step has no analogue; the dihedral and symmetric cases remain open, and the post-quantum standards are built on that ground.

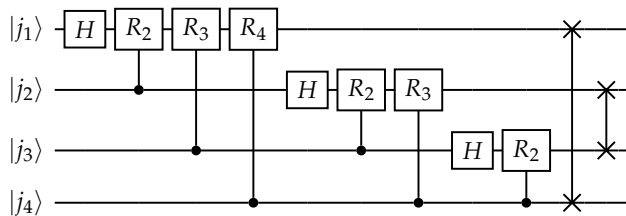
Backup: the QFT circuit

For $Q = 2^t$ and $|j\rangle = |j_1 \cdots j_t\rangle$ big-endian,

$$F_Q |j_1 \cdots j_t\rangle = \frac{1}{2^{t/2}} \bigotimes_{\ell=1}^t \left(|0\rangle + e^{2\pi i(0.j_{t-\ell+1} \cdots j_t)} |1\rangle \right),$$

so the transform of a basis state is a product state and the wires can be prepared independently.

With $R_k = \text{diag}(1, e^{2\pi i/2^k})$:



Cost: t Hadamards, $\binom{t}{2}$ controlled rotations, $\lfloor t/2 \rfloor$ swaps, $\Theta(t^2)$ gates at depth $\Theta(t)$. Dropping every R_k with $k > m$ gives error $\leq 2\pi t/2^m$ (Coppersmith), so $m = O(\log(t/\epsilon))$ and $O(t \log t)$ gates suffice. The inverse is the same circuit reversed with every rotation angle negated.

Backup: reversible modular arithmetic

The controlled multiplier is built in four levels. Each block restores its ancillas, as **P5** requires.

Level	Object	Construction
0	Toffoli, temporary AND	$7T$; $4T$ per compute/uncompute pair
1	ripple-carry adder	MAJ chain, CNOT, UMA chain: $2n$ Toffoli, $2n+2$ qubits, no garbage
2	modular adder mod N	7 blocks: add, subtract N , detect sign, restore, clear flag
3	$\text{CMULT}(a)$	n doubly-controlled modular additions of $2^i a \bmod N$
4	$c\text{-}U_a$	$\text{CMULT}(\bar{a})^\dagger \cdot c\text{-SWAP} \cdot \text{CMULT}(a)$

Level 4 uses $\bar{a} = a^{-1} \bmod N$, computed classically; its existence is what permits an in-place multiplier. For $c = 1$:

$$|x\rangle|0\rangle \rightarrow |x\rangle|ax\rangle \rightarrow |ax\rangle|x\rangle \rightarrow |ax\rangle|x - \bar{a}(ax)\rangle = |ax\rangle|0\rangle.$$

Controlling every gate of a block is wasteful. In the modular adder only 3 of the 7 blocks require the outer control; the remaining four compose to the identity when the control is off, provided the value register satisfies $b < N$, which the reduction maintains. Measured on $N = 15$, $a = 7$: 206 CNOTs against 2634 for the naive construction.

Backup: logical resources across key sizes

Simulated logical counts for the constructions of Roetteler–Naehrig–Svore–Lauter (curves) and Häner–Roetteler–Svore (RSA, minimal width $2n + 2$). Paired rows carry comparable classical security.

ECC bits	qubits	Toffoli	RSA bits	qubits	Toffoli
110	1 014	9.4×10^9	512	1 026	6.4×10^{10}
160	1 466	3.0×10^{10}	1 024	2 050	5.8×10^{11}
224	2 042	8.4×10^{10}	2 048	4 098	5.2×10^{12}
256	2 330	1.3×10^{11}	3 072	6 146	1.9×10^{13}
384	3 484	4.5×10^{11}	7 680	15 362	3.3×10^{14}
521	4 719	1.1×10^{12}	15 360	30 722	2.9×10^{15}

The qubit column is exactly $9n + 2\lceil \log_2 n \rceil + 10$ on every row, and $(448 \log_2 n + 4090)n^3$ reproduces the ECC Toffoli column to a few percent.

Caveats. These are logical counts under a uniformly good hardware model; error correction and magic-state production multiply both sides by comparable factors. The Ekerå–Håstad reduction divides the RSA column by 4 without saving qubits, and applies where the exponent structure is known, so not to a general ECDLP. The windowed $\approx 0.3n^3$ construction of section 8 optimises spacetime volume at $\approx 3n$ qubits and belongs to a different design point, so its figures are not comparable row-for-row with this table. A hybrid quantum–classical sieve using $n^{2/3+o(1)}$ qubits would move the alignment again.

Backup: the elliptic-curve oracle in detail

Semiclassical variant. Every control of the two transforms is a qubit measured at the end and never reused, so each index register collapses to a single recycled ancilla driven by classical feedback. This saves $2n$ qubits at no gate cost, and the same argument applies to the counting register of section 5.

Block costs (Roetteler–Naehrig–Svore–Lauter, ASIACRYPT 2017; $n = \lceil \log_2 p \rceil$).

Circuit	Qubits	Ancillas	Toffoli
<code>ctrl_add_const_modp</code>	$2n+1$	n	$16n \log_2 n - 26.9n$
<code>mul_modp</code> (Montgomery)	$5n+4$	$2n+4$	$16n^2 \log_2 n - 26.3n^2$
<code>sq_modp</code> (Montgomery)	$4n+5$	$2n+5$	$16n^2 \log_2 n - 26.3n^2$
<code>inv_modp</code> (Kaliski)	$7n+2\lceil \log_2 n \rceil+9$	$5n+2\lceil \log_2 n \rceil+9$	$32n^2 \log_2 n$
controlled point addition	$9n+2\lceil \log_2 n \rceil+10$	—	$\lesssim 224n^2 \log_2 n + 2045n^2$
full run, $2n$ additions	$9n+2\lceil \log_2 n \rceil+10$	—	$(448 \log_2 n + 4090) n^3$

Kaliski’s inversion has a data-dependent loop length $n \leq k \leq 2n$, so the circuit runs a fixed $2n$ rounds with a flag that freezes the state once $v = 0$, and one qubit per round records the branch — which suffices because exactly one of the two residues is even at each step. Those $2n$ records dominate the width: the peak occurs inside an inversion, at $7n + 2\lceil \log_2 n \rceil + 9$ plus one control plus $2n$ for the coordinate registers. Uncomputing λ uses the collinearity of P_1, P_2 and $(x_3, -y_3)$, so $\lambda = -(y_3 + y_2)/(x_3 - x_2)$ is recomputable from the output. The precondition $(x_1, y_1) \neq \pm P_2$, which keeps Δx invertible, is discharged in the literature only up to a measure-zero fraction of branches and is the least rigorous part of the published analysis.

Backup: precision and the choice of t

Theorem. Fix $\nu \geq 1$ and $\varepsilon \in (0, \frac{1}{2})$, and run phase estimation with

$$t = \nu + \lceil \log_2 (2 + \frac{1}{2\varepsilon}) \rceil$$

counting qubits. Then $|\ell/Q - \theta| < 2^{-\nu}$ with probability $\geq 1 - \varepsilon$.

Sketch. $|\alpha_\ell| \leq \frac{1}{2|Q\delta - \ell|}$ from $|1 - e^{i\vartheta}| \geq 2|\vartheta|/\pi$; summing the tail beyond $e = 2^{t-\nu} - 1$ gives $\Pr \leq \frac{1}{2(e-1)}$, and $|\ell| \leq e$ implies accuracy $2^{-\nu}$.

Two admissible choices for order finding:

- $t = \lceil 2 \log_2 N \rceil$, so $N^2 \leq Q < 4N^2$. The nearest grid point lies within $1/(2Q) \leq 1/(2N^2)$ and is obtained with probability $\geq 4/\pi^2$ per branch.
- $t = 2n + 1 + \lceil \log_2 (2 + \frac{1}{2\varepsilon}) \rceil$, giving accuracy $2^{-(2n+1)}$ with probability $\geq 1 - \varepsilon$.

Both give $t = 2n + O(1)$. The additional qubits purchase confidence rather than resolution.

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